

Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Online Goods and Services Mock Examination

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Total Marks: 150 Marks

Question 1: Classification & Features of Online Services (Total: 40 marks)

Making services available online has fundamentally shifted accessibility, enabling users to interact with platforms anywhere using any connected device.

(a) State the historical alternative infrastructure used for remote service access before the development of the internet. (1)

Response space

(b) Identify **four** distinct types of data or elements stored inside an online product catalogue. (4)

1.

Element 1

2.

Element 2

3.

Element 3

4.

Element 4

(c) Online retailers use structural hierarchies to systematically organize extensive item lines. Describe how products are grouped and displayed on shopping sites to streamline user navigation. (2)

Description space

(d) Review the checkout features of commercial retail platforms.

(i) State the function of the online shopping basket. (1)

Function text

(ii) State what feature a customer can utilize at checkout to receive a monetary discount on their final balance. (1)

Feature text

(iii) State the policy standard regarding user profiling that allows a consumer to check out without registering an account. (1)

Policy standard text

(e) Digital booking setups manage logistics, seating allocation, and dispatch protocols wirelessly.

(i) List **three** choices or features a customer specifies when executing an online travel booking. (3)

Choice/Feature 1

Choice/Feature 2

Choice/Feature 3

(ii) Identify **two** delivery mechanisms used by ticketing platforms to issue valid passes to a user's smartphone. (2)

Delivery Mechanism 1

Delivery Mechanism 2

(f) Financial institutions package complex banking applications into secure digital interfaces. Identify **four** online services explicitly provided by modern online banks. (4)

Online Banking Service 1

Online Banking Service 2

Online Banking Service 3

Online Banking Service 4

(g) Complete the following comparison table matching specialized service providers with their core functionalities. (6)

Online Service Category	Core Intended Operational Focus / Target Materials	Sourced Platform Manifestation
Education Providers	(i)	Virtual Learning Environments (VLEs)
Gaming Sites	(ii)	Hosted remote server multiplayer suites
Auction Sites	(iii)	Bidding windows and fixed-price listings

(h) News and informational networks rely on background synchronization to push updates to their audiences. State the specific feature of smartphone applications used by news services to send real-time alerts about weather or stock markets directly to a user's display screen. (1)

Response space

(i) Peer-to-peer auction platforms require built-in verification models to build transaction safety. Explain why an auction platform must provide a formal rating system for buyers and a separate rating system for sellers. (4)

Buyer Ratings:

Explanation for buyer ratings

Seller Ratings:

Explanation for seller ratings

(j) Banks enforce high-tier cryptographic security across all consumer-facing portals. Explain why online banking operations must be conducted over a secure HTTPS framework. (3)

Explanation space

(k) Identify **one** action from the list below that banks do *not* typically provide as a direct online service. Place a cross (x) in the correct box. (1)

- ☐ A. Transferring funds between foreign accounts
- ☐ B. Viewing a live ledger account balance
- ☐ C. Paying in physical cash or paper cheques
- ☐ D. Applying for a corporate financial loan

(l) State **two** ways in which a digital education provider can use its VLE utilities to support remote learners. (2)

Method 1

Method 2

(m) Explain the operational or security benefit of utilizing an e-ticket linked directly to a customer's passport for international travel. (2)

Explanation space

Question 2: Impact on Lifestyle, Health, and Behavior (Total: 30 marks)

Shifting core social, educational, and commercial processes to the internet alters cognitive attention spans, physical agility, and community structures.

(a) Define the overarching positive impact that online goods and services introduce regarding human time management. (2)

Definition text

(b) Automated consumer applications remove significant friction from routine household tasks.

(i) Identify **three** distinct lifestyle benefits derived specifically from utilizing online grocery shopping platforms. (3)

Lifestyle Benefit 1

Lifestyle Benefit 2

Lifestyle Benefit 3

(ii) Explain how an automated rail booking system improves travel convenience for commuters compared to traditional terminal kiosks. (3)

Explanation text

(c) Online banking reduces a consumer's reliance on physical banking branches and hard currency. State why bypassing the need to physically carry hard cash establishes a safer personal lifestyle for the individual. (2)

Response space

(d) Automated traffic advisors update routing vectors dynamically based on field measurements. Describe how online information providers interact with GPS devices to optimize travel times for motorists during active delays. (2)

Description text

(e) Complete the comprehensive summary table below by directly entering the target positive lifestyle markers or negative behavioural risks detailed in the textbook guidelines. (8)

Structural Evaluation Domain	Sourced Lifestyle Impact / Behavioral Marker	Sourced Target Manifestation Example
Positive Lifestyle Summary	• Maximizes quality time spent with family and friends. • Minimizes safety hazards by removing cash requirements. • (i) <i>[Response Space]</i>	Heightened individual engagement with creative media assets (e.g., photography, art, videos).

Negative Behavioural Risks	• (ii) <i>[Response Space]</i> • Prompts severe social detachment and community fragmentation. • (iii) <i>[Response Space]</i> • (iv) <i>[Response Space]</i>	Immediate service access reduces personal patience and increases demands on others.
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(f) Constant exposure to bite-sized internet text arrays alters how human cognitive networks process data streams. Explain the negative impact that a constant habit of skim-reading online information has on an individual's general attention to detail. (3)

Explanation text

(g) Explain why an over-reliance on external digital databases to retrieve everyday reference details can cause a measurable negative impact on human memory capacity. (3)

Explanation text

(h) State the primary vulnerability a modern individual faces when their daily lifestyle becomes completely reliant on online services, specifically during a sudden infrastructure blackout. (2)

Vulnerability response space

Question 3: Impact on Business & Operational Operations (Total: 25 marks)

Integrating public web applications forces commercial entities to restructure their communication matrices, data assets, and competitive stances.

(a) Identify **six** core changes or structural transformations that dictate how modern organizations must adapt and run their day-to-day operations in an internet environment. (6)

Core Structural Change 1

Core Structural Change 2

Core Structural Change 3

Core Structural Change 4

Core Structural Change 5

Core Structural Change 6

(b) Global digital marketplaces eliminate traditional localized trade boundaries.

(i) Describe how an organization leverages internet services to connect with global markets without physical branch locations. (2)

Description text

(ii) Describe how an organization utilizes an internet framework to build a 24-hour business cycle by leveraging a global workforce. (2)

Description text

(c) Large-scale data ingestion allows organizations to discover hidden consumer trends. Identify the technical concept responsible for parsing massive quantities of network data to map consumer behavior profiles, and explain why this presents a security challenge for the organization. (3)

Technical Concept:

Concept name

Security Challenge:

Challenge explanation

(d) The removal of geographical constraints exposes local vendors to aggressive international market pricing. State the negative competitive impact this shifts onto legacy local businesses that fail to innovate. (2)

Response text

(e) Complete the business metrics cross-matching chart below by matching the corporate shift with its correct functional outcome. (4)

- Improved Communication:

Outcome mapping

- Big Data Processing:

Outcome mapping

- Information Security:

Outcome mapping

- Increased Competition:

Outcome mapping

(f) Discuss why expanding an enterprise framework into public remote environments inherently increases the severity of its information security threat landscape. (3)

Discussion text

Question 4: Transactional Data & Tracking Architecture (Total: 30 marks)

Web nodes document active network interactions by writing system parameters to hidden client-side storage files.

(a) State the precise technical definition of an online transaction, clarifying why it is not limited strictly to financial exchanges. (2)

Definition space

(b) Define the term cookie and state its primary functional objective when deployed by an enterprise domain. (2)

Response space

(c) Web browsers cache diverse field inputs inside background cookie text arrays during a transaction loop. Identify **six** unique types of data or interaction steps that can be captured and stored inside client-side cookies. (6)

Cookie Data Type 1

Cookie Data Type 2

Cookie Data Type 3

Cookie Data Type 4

Cookie Data Type 5

Cookie Data Type 6

(d) Complete the following data classification table defining cookie lifetimes and tracking models. (6)

Cookie Classification Category	Sourced Runtime Lifespan / Performance Rule	Sourced Operational Deployment Context
1. Session Cookies	(i) <i>[Response Space]</i>	Cleared instantly upon tab closure to maintain quick state data.
2. Persistent Cookies	Records technical usage profiles across multiple sessions over time.	(ii) <i>[Response Space]</i>
3. Third-Party Cookies	(iii) <i>[Response Space]</i>	Placed via ad servers when rendering embedded banner advertisements.

(e) Define the automated analytical process known as targeted marketing. (2)

Definition text

(f) State one specific consumer privacy drawback or ethical risk caused by the unrestricted deployment of third-party tracking cookies across public domains. (2)

Privacy risk text

(g) Differentiate between a session cookie and a persistent cookie regarding device non-volatility parameters and browser execution loops. (3)

Differentiation text

(h) Describe how an advertising agency can compile a comprehensive user browsing profile by reading cross-site data paths via third-party cookies. (4)

Description space

Question 5: Extended Response Essay — Data Tracking & Individual Impacts (Total: 25 marks)

The pervasive use of text-tracking files and persistent cookies by commercial web organizations creates a profound impact on individual digital privacy and consumer autonomy.

An international e-commerce conglomerate plans to integrate an advanced behavioral tracking core into its global retail portal. The system logs all user interactions via third-party advertising cookies, saves extensive form inputs permanently to persistent cookies, and uses big data analytics to update home streams dynamically with targeted marketing banners.

Evaluate the implications of implementing widespread transactional data tracking systems from the perspective of individual consumers. In your response, you must discuss:

- How the collection of structural transactional parameters (such as form inputs, IP addresses, and basket selections) balances user convenience against personal surveillance concerns.
- The core functional distinctions and privacy impacts of session cookies versus persistent and third-party tracking configurations.
- The downstream consequences that continuous automated targeted marketing profiles impose on individual consumer choice, personal financial habits, and data confidentiality boundaries.

Plan your answer carefully. Marks will be awarded for structured technical progression, precise usage of technical ICT vocabulary from the text (such as persistent cookies, transactional data, big data), balanced arguments, and a clear, well-supported final conclusion.

Student response essay space (25 Marks)

End of Question Bank